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IoT ACE02
Computer Vision

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Short answer test

Assessment tool - 1

1) Computer vision is a branch of AI that enables computers to acquire, process, analyze and understand images and videos. It extracts meaning from information from the visual data and helps computers to make decisions similar to human vision.

a) Main objective of computer vision:-

- * Detect and recognize objects.
- * Interpret scene and images
- * Extract useful information from image.

b) Real world applications:-

- * Face recognition system.
- * Self driving (automation) vehicles.

2) Key features:-

- * Feature extraction.
- * Object detection
- * Object recognition
- * Image segmentation
- * Pattern recognition.

a) Feature extraction is the process of identifying important characteristics such as corners, edges, textures and shape. from the image for simple analysis.

b) object recognition identifies and classifies object in an image and make accurate decisions.

3) Image Processing is divided into 3 types:

they are:-

- 1) low level processing
- 2) mid level processing
- 3) High level processing.

a) low level processing improves image quality through operations such as noise removal, filtering & contrast enhancement.

b) mid level processing

- * segmentation and features extraction.

- * provides image features

High level processing

- * object recognition & scene understanding.

- * produces meaningful interpretations.

4) A digital image is represented as a matrix of pixels, where each pixel store intensity or color information.

a) A pixel is the smallest unit of a digital image that represents brightness or color.

b) Higher resolution contains, more pixels, producing sharper and brighter images, where as low resolution contains less pixels, and low lighting and pixels may miss.

5) Image ~~any~~ acquisition captures light from a real world scene using a image sensors and converts it into electrical signals.

a) A image sensor converts coming light into electrical signals.

b) * digital camera.
* web camera.

6) Sampling converts a continuous image into discrete values. pixels. while quantization assign intensity value to each pixel.

a) Sampling is a process in which the continuous image is divided into discrete pixels.

b) Quantization converts continuous intensity values into discrete digital values for computer processing.

7) The image formation model explains how light from a scene passes through a cam lens and forms a digital image on the image sensor.

a) Perspective projection is the process of projecting a 3D scene onto a 2D image plane while preserving depth perception.

b) Illumination provides the light required for capturing visible details and affect image brightness and quality.

9) a) Image processing is a technique of enhancement of digital image using various algorithms.

b) Image processing

- * enhance image
- * provides improved images

Computer vision

- * understands image content
- * provides meaningful decisions

9) Computer vision is widely used in healthcare, transportation, surveillance, agriculture, and manufacturing.

a) Computer vision assists doctors by detecting diseases from X-rays, MRI scans, and CT images.

b) It detects roads, traffic signs, and obstacles to enable safe autonomous driving.

10) Digital image fundamentals such as pixels, sampling, quantization, and resolution provide the data required for accurate image analysis and object recognition.

a) Pixels are the basic building blocks because every image is made of pixels.

b) sampling determines the ~~no~~ of pixels, while higher resolution improves image analysis accuracy.